

QUERYPILOT LOCAL

# Evidence first. AI explains.

Offline PostgreSQL plan analysis with  
deterministic rules, local RAG and  
grounded AI.



# The diagnosis is complete before the model starts

1. VALIDATE

2. PROVE

3. GROUND

Read-only SQL

Plan evidence

Grounded AI

AST gate + 3 s  
database timeout

Normalize JSON +  
apply deterministic  
rules

Retrieve chunks +  
generate evidence-  
bound prose

# Local AI + RAG extends the deterministic core

## LOCAL RUNTIME

PostgreSQL + FastAPI +  
rules + RAG + Foundry.

## PUBLIC DEMO

Browser-only plan JSON;  
no database or model.

Input

One read-only SELECT

Core

Parser + rule engine

Optional

RAG chunks + grounded AI prose

Public

EXPLAIN JSON in browser

# No evidence, no AI recommendation

Reliability is enforced before and after generation

Category, severity, raw plan evidence and recommendation SQL are application-owned. Generated prose must reference known evidence and retrieved chunk IDs. Invented metrics, identifiers or citations fall back safely.

- ④ Single read-only SELECT
- ④ Plan evidence required
- ④ Retrieved citation IDs only
- ④ Recommendation SQL stays display-only
- ④ No-answer is a valid result

# Grounded AI passes 4/4; CPU latency stays optional

## MEASURED RESULT

12 committed scenarios. Rule and no-answer: 12/12; retrieval Hit@3: 9/9; grounded reports and citations: 4/4.

12/12

Rule diagnosis  
and no-answer

9/9

Retrieval  
Hit@3

48.3 s

Average optional  
CPU generation

## EVIDENCE

# Proof measured. AI grounded.

1.671 ms Seq Scan → 0.074 ms Index Scan (7-run medians)  
Grounded generation 4/4 · valid citations 4/4 · CPU average 48.3 s  
Local fixture only · next: authorized non-production pilot

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